

Lesson 41 Practice: Networking Protocols

JKSSB Computer Awareness

October 10, 2026

Provenance: All 25 questions are original exam-pattern questions (*[Original]* plus the *[JKSSB FAA-pattern]*, *[JKPSI-pattern]*, *[SSC CGL-pattern]*, and *[JKSSB-pattern]* extension items, including MSQs); none reproduces an official paper item.

Q1. *[Original]* Expand TCP.

- A. Transfer Control Program
- B. Transmission Control Protocol
- C. Transport Communication Process
- D. Tele Communication Procedure

Q2. *[Original]* Expand DNS.

- A. Domain Name System
- B. Data Network Service
- C. Digital Number Sequence
- D. Direct Node Setup

Q3. *[Original]* Which pair correctly matches the transport protocol with its behaviour?

- A. TCP — fast, connectionless, no retransmission
- B. UDP — reliable, ordered, connection-oriented
- C. TCP — reliable, ordered, connection-oriented
- D. UDP — ordered delivery with guaranteed timing

Q4. *[Original]* TCP guarantees:

- A. fast arrival time of every packet.
- B. complete and ordered delivery of bytes.
- C. encryption of the message body.
- D. translation of names into addresses.

Q5. *[Original]* Which of the following is **NOT** true of UDP?

- A. It is connectionless.
- B. It offers best-effort delivery without retransmission.
- C. It guarantees ordered delivery of all datagrams.

D. It is preferred for streaming and DNS queries.

Q6. *[Original]* IPv4 uses _____ addresses, whereas IPv6 uses _____ addresses.

- A. 128-bit; 32-bit
- B. 32-bit; 128-bit
- C. 64-bit; 256-bit
- D. 32-bit; 256-bit

Q7. *[Original]* Which statement about IPv6 is correct?

- A. IPv6 addresses are 256-bit long.
- B. IPv6 uses four decimal octets.
- C. IPv6 addresses are 128-bit long with a fixed 40-byte header.
- D. IPv6 headers vary between 20 and 60 bytes.

Q8. *[Original]* ICMP is best described as:

- A. a file-transfer protocol with its own port number.
- B. a control and diagnostic protocol carrying errors and probes.
- C. a mail-sending protocol for outgoing messages.
- D. an encryption layer for secure web pages.

Q9. *[Original]* Which tool pair belongs to ICMP?

- A. ping and traceroute
- B. mail merge and autosum
- C. FTP and Telnet
- D. ping and mail merge

Q10. *[Original]* The service that translates `jkssb.nic.in` into an IP address is:

- A. DHCP
- B. ICMP
- C. DNS
- D. SMTP

Q11. *[Original]* A newly connected office desktop receives its IP address, mask, gateway, and DNS server automatically. This service is:

- A. DNS
- B. DHCP
- C. ICMP
- D. FTP

Q12. *[Original]* Which of the following is correctly matched?

- A. HTTP — port 443, encrypted with SSL/TLS

- B. HTTPS — port 80, plain text
- C. HTTP — port 80, plain text
- D. HTTPS — port 25, used for sending mail

Q13. *[Original]* SMTP is used to:

- A. send outgoing mail to the server.
- B. download received mail to the local machine.
- C. keep mail synced on the server.
- D. translate a domain name into an address.

Q14. *[Original]* Which pairing correctly matches the mail protocol with its role?

- A. SMTP — receives and syncs mail on the server
- B. POP3 — downloads received mail to the local machine
- C. IMAP — sends outgoing mail to the server
- D. SMTP — translates names into IP addresses

Q15. *[Original]* Evaluate the statements:

- I. TCP is connection-oriented and delivers bytes in order.
 - II. IPv6 addresses are 128-bit long.
 - III. ICMP transfers application data over a dedicated port.
- A. I only
 - B. I and II only
 - C. II and III only
 - D. I, II, and III

Q16. *[Original]* Which layer pair is correctly matched in the OSI model?

- A. TCP and UDP — Network layer
- B. IP and ICMP — Transport layer
- C. HTTP, FTP, SMTP, DNS — Application layer
- D. IP and HTTP — Physical layer

Q17. *[Original]* In the OSI model, the correct order from layer 1 to layer 7 is:

- A. Application, Presentation, Session, Transport, Network, Data Link, Physical
- B. Physical, Data Link, Network, Transport, Session, Presentation, Application
- C. Physical, Network, Data Link, Session, Transport, Application, Presentation
- D. Data Link, Physical, Transport, Network, Application, Session, Presentation

Q18. *[Original]* Which port pairing is correctly matched?

- A. FTP — 23; Telnet — 21

- B. SMTP — 25; DNS — 53
- C. HTTP — 443; HTTPS — 80
- D. POP3 — 143; IMAP — 110

Q19. *[Original]* Which of the following is **INCORRECT**?

- A. FTP moves files between computers (port 21).
- B. Telnet opens a plain-text remote login session (port 23).
- C. DHCP assigns addresses to devices joining the network.
- D. ICMP uses port 80 to transfer web pages.

Q20. *[Original]* An office PC cannot open any website, but a ping to the gateway succeeds. Which conclusion follows?

- A. The network path works, so the fault lies above it (DNS, proxy, or application settings).
- B. The ping proves the web server has sent the page.
- C. ICMP has delivered the web page to the browser.
- D. The Translate layer of the OSI model has failed.

Q21. *[JKSSB FAA-pattern]* TCP/IP is best described as:

- A. the core protocol suite of the Internet.
- B. a printer ink formula.
- C. a type of monitor glass.
- D. a keyboard layout.

Q22. *[JKPSI-pattern]* An IP address identifies a:

- A. device on a network.
- B. printed page number.
- C. speaker's volume.
- D. book's author.

Q23. *[SSC CGL-pattern]* IPv4 addresses are how many bits long?

- A. 32
- B. 8
- C. 128
- D. 256

Q24. *[JKSSB-pattern]* (**MSQ: select ALL correct options.**) Which are network protocols?

- A. TCP/IP
- B. HTTP
- C. FTP
- D. Stapler pin

- Q25.** *[SSC CGL-pattern]* (MSQ: select ALL correct options.) Which statements about protocols are correct?
- A. TCP gives reliable ordered delivery.
 - B. IP handles addressing and routing.
 - C. FTP transfers files between systems.
 - D. An IP address identifies a printed page.
- Q26.** *[PYQ-adapted: JKSSB JKP Constable 2024, QCRIAS item 747; SRC-02]* In HTTPS, what does the letter S stand for?
- A. Selected
 - B. Secure
 - C. Software
 - D. System
- Q27.** *[PYQ-adapted: JKSSB JKP Constable 2024, QCRIAS item 765; SRC-02]* Which organization develops standards for the World Wide Web?
- A. W3C
 - B. WHO
 - C. OPEC
 - D. UNESCO
- Q28.** *[Original]* Which protocol provides reliable, ordered delivery between applications?
- A. TCP
 - B. UDP
 - C. ARP
 - D. ICMP
- Q29.** *[Original]* Which protocol translates a domain name into an IP address?
- A. DNS
 - B. SMTP
 - C. FTP
 - D. POP3
- Q30.** *[Original, MSQ drill]* (MSQ: select ALL correct options.) Which are application-layer protocols/services?
- A. HTTP
 - B. SMTP
 - C. FTP
 - D. Ethernet framing